



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

MA1502-1 – NUMBER THEORY

(Open Elective)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. For $a = 20$ and $b = 7$, the division algorithm gives
A) $q = 2, r = 3$ B) $q = 2, r = 6$
C) $q = 2, r = 1$ D) $q = 3, r = 0$
2. If $a|b$ and $b|a$ then
A) $a = b$ B) $a = \pm b$
C) $a = b = 0$ D) $a = \pm 1$
3. Let $\gcd(a, b) = d$, then
A) $d|a$ only B) $d|b$ only
C) $d|a$ and $d|b$ D) $a = b = d$
4. The inverse of 3 modulo 7 is?
A) -1 B) -2
C) -3 D) -4
5. The solution of the linear congruence $4x \equiv 5 \pmod{9}$ is?
A) $6 \pmod{9}$ B) $8 \pmod{9}$
C) $9 \pmod{9}$ D) $10 \pmod{9}$
6. The prime factorization of 1001 is
A) $7^3 \cdot 11 \cdot 13$ B) $7^2 \cdot 11 \cdot 13$
C) $7 \cdot 11 \cdot 13$ D) $7 \cdot 11^3 \cdot 13$
7. The linear congruence $19x \equiv 6 \pmod{30}$ has
A) Infinitely many solutions modulo 30 B) Unique solution modulo 30
C) Exactly 6 solution modulo 30 D) none
8. For the equation $30x + 42y = 12$, which value of x, y is valid?
A) $x = 1, y = -1$ B) $x = 0, y = 1$
C) $x = -1, y = 1$ D) $x = -2, y = 1$
9. If p is a prime, then
A) $p - 1! \equiv 1 \pmod{p}$ B) $p - 1! \equiv -1 \pmod{p}$
C) $p - 1! \equiv 0 \pmod{p}$ D) $p! \equiv 0 \pmod{p}$
10. $\phi(13)$ is:
A) 13 B) 12
C) 14 D) 11
11. Remainder of 2^{50} when divided by 7 is
A) 4 B) 5
C) 0 D) 1

12. Remainder of $15!$ when divided by 17 is
 A) -1 B) 1
 C) 3 D) 2
13. Order of $2 \bmod 13$ is
 A) 2 B) 3
 C) 6 D) 12
14. How many primitive roots are there for 25?
 A) 4 B) 5
 C) 7 D) 8
15. When $p=13$, using Legendre symbol $4/13$ is equal to
 A) 1 B) -1
 C) 0 D) 2
16. If $a \equiv b \pmod{n}$ then for any integer k
 A) $a^k \equiv b^k \pmod{n}$ B) $a^k \equiv b \pmod{n}$
 C) $a \equiv b \pmod{n}$ D) $a \equiv 0 \pmod{n}$
17. Which of the following is NOT a key pair in public key cryptography?
 A) Public key and private key. B) Symmetric key and public key.
 C) Encryption key and decryption key. D) Private key and decryption key.
18. The RSA encryption function is:

$$C = M^e \bmod n$$

Here, e represents:

- A) The private key exponent B) The public key exponent
 C) The modulus D) The message length
19. RSA is considered secure because
 A) Factoring large integers is computationally hard B) Modular exponentiation is inefficient
 C) It uses symmetric keys D) It works only for small primes
20. The Diffie-Hellman key exchange uses primitive roots to
 A) Generate symmetric keys B) Allow secure key exchange over insecure channels
 C) Factor large integers D) Encrypt messages directly

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Prove that the integers $53^{103} + 103^{53}$ is divisible by 39.	6	L*2	2	2
b) Solve $x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$	6	L2	1	1
c) Let a, b, c, d be any nonzero integers. Prove that (i) If $a \mid b$ and $a \mid c$, then $a^2 \mid bc$ (ii) If $a \mid b$ and $c \mid d$, then $ac \mid bd$.	4	L3	1	1
2. a) Use the Euclidean algorithm to obtain integers x & y satisfying $\gcd(254, 32) = 254x + 32y$.	6	L2	1	1
b) i) If $ca \equiv cb \pmod{n}$ then prove that $a \equiv b \pmod{\frac{n}{d}}$ where $d = \gcd(c, n)$. ii) What is the remainder of $1! + 2! + 3! + \dots + 100!$ Divided by 24.	6	L2	2	1
c) Solve the linear congruence $25x \equiv 15 \pmod{29}$.	4	L3	2	1
3. a) Determine all solutions of the Diophantine equation $56x + 72y = 40$	6	L2	1	2
b) Use theory of congruence to verify that $89 \mid 2^{44} - 1$ and $97 \mid 2^{48} - 1$.	6	L3	2	1
c) Solve $7x + 3y \equiv 6 \pmod{11}, 4x + 2y \equiv 9 \pmod{11}$.	4	L2	1	2

Unit – II

- | | | | | | |
|-------|--|---|----|---|---|
| 4. a) | State and prove Fermat's Little theorem. | 6 | L2 | 3 | 2 |
| b) | If n is an odd integer, prove that $\varphi(2n) = \varphi(n)$ and If n is an even integer, prove that $\varphi(2n) = 2\varphi(n)$. | 6 | L2 | 3 | 1 |
| c) | Let p be an odd prime and $\gcd(a, p) = 1$, use Legendre symbol $\left(\frac{a}{p}\right)$ to find (i) $\left(\frac{18}{43}\right)$ (ii) $\left(\frac{19}{23}\right)$ | 4 | L3 | 4 | 1 |
| 5. a) | Use Fermat's theorem to verify that 17 divides $11^{104} + 1$ | 6 | L2 | 3 | 1 |
| b) | Prove that for $n > 2$, $\varphi(n)$ is an even integer. | 6 | L2 | 3 | 2 |
| c) | Solve the quadratic congruences $x^2 + 7x + 10 \equiv 0 \pmod{11}$. | 4 | L2 | 4 | 1 |
| 6. a) | Prove that for any integer a have order k modulo n . Then $a^h \equiv 1 \pmod{n}$ if and only if $k h$. | 6 | L2 | 4 | 2 |
| b) | If p is a prime and $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ $a_n \not\equiv 0 \pmod{p}$ is a polynomial of degree $n \geq 1$ with integral coefficients, then prove that the congruence $f(x) \equiv 0 \pmod{p}$ has at most n incongruent solutions modulo p . | 6 | L3 | 4 | 2 |
| c) | If p and q are distinct primes, prove that $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$. | 4 | L2 | 3 | 2 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|---|
| 7. a) | Use the Hill cipher | | | | |
| | $C_1 \equiv 2P_1 + 3P_2 \pmod{26}$ $C_2 \equiv 5P_1 + 8P_2 \pmod{26}$ to encipher the message BUY NOW. | 8 | L2 | 5 | 1 |
| b) | Encipher the message HAVE A NICE TRIP using a Vigenère cipher with the keyword MATH. | 8 | L3 | 5 | 2 |
| 8. a) | Using linear cipher $C \equiv 5P + 11 \pmod{26}$, encrypt the message NUMBER THEORY IS EASY. | 8 | L2 | 5 | 1 |
| b) | Encrypt the plaintext message GOLD MEDAL using the RSA algorithm with key (2561, 3). | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
